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HEAS: A HIERARCHICAL EVOLUTIONARY AGENT-BASED SIMULATION FRAMEWORK FOR MULTI-OBJECTIVE POLICY SEARCH

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ABSTRACT

Agent-based frameworks provide no native support for the ODE-style policy-search pipeline that simulation studies demand: multi-objective evolutionary search, multi-scenario tournament comparison, and statistically rigorous multi-run validation. Researchers bridge this gap with bespoke coupling code—100–200 lines per project—that is non-reusable and prone to silent metric divergence. HEAS (Hierarchical Evolutionary Agent-based Simulation) addresses this through four commitments: a Layer/Stream/Arena composition model; a uniform metric contract consumed identically by the evolutionary algorithm, tournament scorer, and bootstrap CI pipeline; native NSGA-II integration; and a Tournament primitive with four voting rules and noise stability analysis. Against a Mesa 3.3.1 baseline, HEAS requires 5 coupling lines versus 160 (97% reduction). Three case studies—ecological, enterprise, and a Wolf-Sheep mean-field reimplementation—are validated at two simulation scales. An evolved champion wins all 32 held-out scenarios, confirming out-of-distribution robustness.

1 INTRODUCTION

160 lines of coupling code—that is the overhead a competent Mesa programmer writes to connect a predator-prey simulation to NSGA-II, tournament comparison, and bootstrap confidence intervals. HEAS reduces this to five lines. The rest of this paper explains why the gap is structural, not stylistic.

Agent-based models (ABMs) have become a primary tool for studying emergent behavior in complex adaptive systems—from ecological communities and financial markets to public health interventions and regulatory policy design. Using an ODE-style or tabular ABM for *policy search* requires integration infrastructure that no general-purpose ABM framework provides as a reusable layer. HEAS targets this specific niche—coupled ODE/tabular dynamics, multi-objective EA, multi-run statistics—rather than the full ABM space including spatially-explicit heterogeneous-agent systems.

1.1 The Coupling Code Problem

Consider a researcher using Mesa to study regulatory policy in a multi-firm economy. She wants to (1) evolve a Pareto-optimal tax regime using NSGA-II, (2) compare the evolved regime against a reference policy across independent economic scenarios using tournament voting, and (3) report bootstrap confidence intervals. Each step reads the same welfare metric, yet in Mesa each is an independent code path: the DEAP fitness function extracts welfare from the model's `DataCollector` after each run; the tournament scorer extracts welfare from a separate episode-evaluation function, possibly with different aggregation; and the bootstrap CI loop extracts welfare from a third data collection loop, potentially inconsistent with both.

When the researcher adds a second objective—welfare inequality—she must edit the DataCollector reporter, the DEAP fitness function, the tournament scorer, and the CI loop, keeping all four in sync manually. A silent divergence can persist undetected for weeks and invalidate a publication. This pattern, which we call *coupling code*, appears in representative ABM-based policy-search workflows (Pangallo, Farmer, Sanders, and Galla 2019; Caprì, Ignaccolo, Inturri, Pluchino, and Rapisarda 2023).

1.2 Contributions

We present **HEAS** through four architectural commitments. **C1—Layered Composition** formalizes multi-level hierarchy through a `Layer`/`Stream`/`Arena` triad: Layers declare their own state and step logic, Streams declare how they couple, and the Arena is visible to the framework, enabling independent testing and structural reuse. **C2—Uniform Metric Contract** requires every `Layer` to implement `metrics_episode()`, returning a `dict[str, float]` of namespaced keys—the *single source of truth* consumed identically by the EA fitness function, tournament scorer, and statistical CI pipeline.

C3—Native Multi-Objective EA wraps DEAP's NSGA-II (Fortin, De Rainville, Gardner, Parizeau, and Gagné 2012; Deb, Pratap, Agarwal, and Meyarivan 2002) with per-run seed management, parallel episode evaluation, and hypervolume tracking; the optimizer is replaceable without modifying simulation or metric code. **C4—Tournament Evaluation** implements four voting rules, sample complexity certification, and noise stability analysis as framework-level primitives available to any Arena.

Section 5 reports 97% coupling code reduction, two-scale reproducibility (30 runs at 150 steps; 20 runs at 1,000 steps), and out-of-distribution champion robustness across 32 held-out ecological scenarios.

2 RELATED WORK

Agent-based modeling has become a primary methodology for studying complex adaptive systems, with applications spanning ecology (Grimm, Berger, Bastiansen, Eliassen, Ginot, Giske, Railsback, and DeAngelis 2006), agent-based computational economics (Tesfatsion 2006), and organizational behavior (Bonabeau 2002). Macal and North (Macal and North 2010) survey the ABM tool landscape and identify a recurring architectural division: frameworks such as Mesa (Kazil, Masad, and Crooks 2020), NetLogo (Wilensky 1999), Repast4Py (Collier and North 2022), and ABIDES (Byrd, Hybinette, and Balch 2020) excel at model execution but provide no native policy-search infrastructure. The ODD protocol (Grimm, Berger, Bastiansen, Eliassen, Ginot, Giske, Railsback, and DeAngelis 2006) advances ABM reproducibility through standardized model description, but does not address the statistical machinery needed for multi-run policy comparison.

Simulation optimization has a rich tradition at WSC (Fu 2002; Hong and Nelson 2009). Fu (Fu 2002) identifies the persistent theory-practice gap between optimization formulations and the coupling code required to connect solvers to simulation models. Hong and Nelson (Hong and Nelson 2009) provide a practitioner-oriented introduction that highlights the per-project engineering burden. The SimOpt testbed (Pasupathy and Henderson 2011) catalogs benchmark problems, while commercial platforms such as OptQuest (OptTek Systems, Inc. 2007) and Pagmo (Biscani and Izzo 2020) supply search backends—yet none address the metric-contract problem specific to multi-objective ABM workflows. Multi-objective simulation optimization has received growing WSC attention (Hunter and Pasupathy 2022), and evolutionary approaches including NSGA-II (Deb, Pratap, Agarwal, and Meyarivan 2002) are increasingly deployed in agent-based settings (Caprì, Ignaccolo, Inturri, Pluchino, and Rapisarda 2023).

Prior ABM policy-search studies in economics (Tesfatsion 2006; Zheng, Trott, Srinivasa, Parkes, and Socher 2022) and complex systems (Pangallo, Farmer, Sanders, and Galla 2019) bridge the framework-optimizer gap with bespoke coupling code. Pangallo et al. (Pangallo, Farmer, Sanders, and Galla 2019) demonstrate the fragility of single-run ABM policy conclusions, motivating the multi-run statistical approach that HEAS formalizes. Ranking and selection methods from the simulation literature (Goldsman and Nelson 1994) address experiment design but assume single-objective outcomes and hand-crafted integration layers. Optimization libraries such as DEAP (Fortin, De Rainville, Gardner, Parizeau, and Gagné 2012), Optuna (Akiba, Sano, Yanase, Ohta, and Koyama 2019), and Pagmo (Biscani and Izzo 2020) provide flexible search backends but define no metric interface shared across optimization, tournament evaluation, and statistical inference. Optuna's objective function is a single callable returning a scalar or tuple: it has no concept of a tournament scorer or bootstrap CI pipeline, so the researcher must still synchronize metric

extraction across consumers manually. HEAS's `metrics_episode()` contract closes this gap by being consumed identically at all three stages.

3 THE HEAS FRAMEWORK

HEAS is implemented in Python 3.11 and depends on DEAP for evolutionary algorithms, NumPy for numerical computation, and SciPy for statistical utilities. All components—EA algorithm, voting rule, statistical estimator—can be replaced without changing simulation or metric contract code.

3.1 Layered Composition

The fundamental abstraction is a `Layer` with four methods (`step`, `metrics_step`, `metrics_episode`, `reset`). All downstream components read the same `metrics_episode()` keys, making the metric dictionary the single source of truth.

Layers compose into **Streams**—named groups forming a coherent sub-simulation—and Streams compose into an **Arena**, which orchestrates step ordering, inter-stream signal routing, and episode-level aggregation. The naming convention `stream_name.metric_name` ensures globally unambiguous keys regardless of Arena size. Explicit composition boundaries make components testable and replaceable without rewriting optimization or tournament code; the uniform metric contract (C2) remains stable as model internals evolve.

3.2 Native Multi-Objective EA Integration

The wrapper handles DEAP's creator/toolbox registration, crossover and mutation wiring, hall-of-fame management, and hypervolume computation. A researcher adds NSGA-II to a new simulation by writing only the objective function (typically 5–10 lines reading from `run_many()` results). The EA algorithm is replaceable: `run_ea()` also accepts CMA-ES and hillclimbing. Section 5.5 shows that at 1,000-step ecological scale, NSGA-II's advantage over random search is large ($d = 1.39$), confirming that algorithm-agnosticism and algorithm quality are complementary properties.

`run_many()` executes n episodes in parallel via `ProcessPoolExecutor`, with per-episode seeds derived deterministically from a base seed, making individual fitness calls reproducible.

3.3 Tournament Evaluation

Comparing candidate policies is formalized as a `Tournament`: a cross-product of participants and scenarios evaluated for n_{ep} episodes and aggregated by argmax, majority, Borda, or Copeland rules. The rules diverge when the competitive structure is non-trivial: argmax is vulnerable to outlier scenarios; majority can cycle; Borda weights magnitudes; Copeland is manipulation-resistant. Universal agreement (as in Section 5.4) indicates genuine dominance; disagreement diagnoses the competitive structure. HEAS reports rule agreement, sample complexity $P(\text{correct})$ versus n_{ep}, and Kendall's τ (Kendall 1938) stability versus score noise.

3.4 Statistical Infrastructure

`heas/utils/stats.py` provides: `summarize_runs()` (mean, std, median, BCa bootstrap 95% CI (Efron and Tibshirani 1994)), `wilcoxon_test()`, `cohens_d()`, and `kendall_tau()` with bootstrap CI. `heas/utils/pareto.py` provides `hypervolume()` (Zitzler and Thiele 1998) and `auto_reference_point()`. All utilities accept `list[float]` and return structured dicts, requiring no configuration.

4 CASE STUDIES

Three case studies span ecological, economic, and published-model domains.

4.1 Ecological Population Management

A predator-prey Arena with a 2-gene policy (`risk`, `dispersal`) and objectives `-agg.mean_biomass`, `agg.cv`. Five streams model climate forcing, landscape quality, prey logistic growth, predator dynamics, and cross-stream aggregation; eight fragmentation×shock scenarios form the evaluation grid. Section 5.3 validates this model at two simulation scales; the evolved champion is subjected to a 32-scenario out-of-distribution robustness test in Section 5.4.

4.2 Enterprise Regulatory Design

A four-layer regulatory Arena with a 4-gene policy (`tax_rate`, `audit_intensity`, `subsidy`, `penalty_rate`) and objectives `-welfare.total`, `welfare.gini`. Firms maximize per-step profit against a stochastic demand schedule; the regulator layer adjusts enforcement intensity each step; the government layer collects tax revenue and redistributes subsidies; the welfare layer aggregates per-firm consumer surplus and computes the Gini coefficient over the terminal distribution. The Pareto trade-off is between aggregate welfare and distributional equality: high subsidies increase total welfare but compress the Gini; high audit intensity improves compliance but reduces firm surplus. The study evaluates 32 scenarios (governance regime × demand shock × audit frequency × firm count × cost structure). Section 5.6 reports cross-domain validation at large scale.

4.3 Wolf-Sheep Predation: Mean-Field Reimplementation

To demonstrate compatibility with *existing published models*, we re-implement the canonical Mesa Wolf-Sheep Predation model (Wilensky 1997; Kazil, Masad, and Crooks 2020)—the flagship example distributed with the Mesa library—as a mean-field ODE Arena. Its published dynamics are Lotka-Volterra (Lotka 1925; Volterra 1926) predator-prey with grass regrowth on a 20×20 spatial grid.

HEAS reimplementation. We implement the mean-field ODE approximation as a 4-layer Arena, deriving each term directly from Mesa's energy mechanics. All six published parameters are used verbatim.

Grass. Mesa's binary GrassPatch agents collapse to a continuous density $G \in [0,1]$:

$$\Delta G = \frac{1-G}{\tau_g} - \frac{SG\gamma}{N}, \tag{1}$$

where $\tau_g = 30$, $N = 400$, and $\gamma \in [0,1]$ is the policy grazing-rate gene.

Sheep ($g_s = 4$, $r_s = 0.04$):

$$\Delta S = \Big(r_s - \frac{\max(0,\, 1 - g_s G\gamma)}{g_s}\Big) S \;-\; \frac{WS}{N}. \tag{2}$$

Wolves ($g_w = 20$, $r_w = 0.05$; h = harvest-rate gene):

$$\Delta W = r_w \frac{WS}{N} \;-\; \frac{\max(0,\, 1 - g_w S/N)}{g_w} W \;-\; hW. \tag{3}$$

Every coefficient maps directly to a published Mesa default; no parameters are ad-hoc. Concretely: `grass_regrowth_time` $\to \tau_g$, `sheep_gain_from_food` $\to g_s$, `sheep_reproduce` $\to r_s$, `wolf_gain_from_food` $\to g_w$, `wolf_reproduce` $\to r_w$, `initial_population` $\to N$ (density normalizer). The spatial grid is collapsed to a density field, an acknowledged simplification (Section 6). Adding EA optimization to this published model required zero additional coupling code: the 8-line objective function reads `metrics_episode()` keys, identical in structure to the other two case studies.

A 30-run algorithm comparison on this model is reported in Section 5.5, where the landscape structure determines which optimizer wins.

5 EVALUATION

5.1 Evaluation Questions and Protocol

We pre-specify five evaluation questions: **(RQ1)** Does HEAS reduce integration overhead versus a competent Mesa implementation? **(RQ2)** Are optimization outcomes reproducible across simulation scales? **(RQ3)** Are tournament rankings stable and is the evolved champion robust out-of-distribution? **(RQ4)** Does the framework remain useful when NSGA-II is not the best optimizer, and how sensitive are results to simulation scale? **(RQ5)** Does HEAS generalize across heterogeneous domains?

All statistics use 95% BCa bootstrap confidence intervals, Wilcoxon signed-rank tests, and Cohen's d effect sizes. Hypervolume is computed against a pooled reference point per comparison group.

5.2 Mesa Comparison: Coupling Code Reduction

We implement the ecological case study as a competent Mesa 3.3.1 model and count non-blank, non-comment coupling code (model dynamics excluded). Measured totals: 160 LOC (Mesa) vs. 5 LOC (HEAS)—155 lines saved (97%).

Table 1: Coupling code: Mesa 3.3.1 vs. HEAS vs. Mesa+utility-library (^{est} = estimated; Mesa from `mesa_eco.py`).

Task	Mesa	Mesa+Util	HEAS
Metric contract / DataCollector setup	15	5^{est}	0
Episode metric extraction	20	5^{est}	0
Sequential episode runner	22	3^{est}	1
Parallel runner (`ProcessPoolExecutor`)	20	5^{est}	0
EA fitness glue	14	5^{est}	3
Tournament scorer	18	5^{est}	0
Per-run seed management (30-run)	12	3^{est}	0
Bootstrap CI	35	5^{est}	0
Adding a second objective (incremental)	4	2^{est}	1
Total	**160**	**~38^{est}**	**5**

Listing 1 reproduces the two most illustrative Mesa coupling blocks verbatim, allowing verification without repository access. Adding a metric requires editing Block A *and* every downstream consumer; in HEAS this is a one-line change in one file.

Listing 1: Mesa coupling code Blocks A and B (35 LOC from `experiments/mesa_eco.py`). HEAS equivalent: zero LOC.

```
# Block A - DataCollector setup (15 LOC)
# Adding a metric: edit here AND the EA function
# AND the tournament scorer, all in sync.
self.datacollector = mesa.DataCollector(
   model_reporters={
      "prey":   lambda m: m.prey,
      "pred":   lambda m: m.pred,
      "extinct": lambda m: float(m.pred <= 0.0),
      "biomass": lambda m: m.prey + m.pred,
   }
)

# Block B - episode metric extraction (20 LOC)
```

```
# Every consumer (EA, tournament, CI) calls this
# separately. Silent divergence risk.
def extract_episode_metrics(model):
    df = model.datacollector.get_model_vars_dataframe()
    mean_biomass = float(df["biomass"].mean())
    cv = df["biomass"].std() / mean_biomass \
        if mean_biomass > 0 else 0.0
    return {"mean_biomass": mean_biomass, "cv": cv,
            "extinct": float(df["extinct"].iloc[-1])}
```

Table 1 includes a Mesa+utility-library estimate (∼38 LOC). HEAS's claim over this baseline is that C2 enforces the metric interface at the *framework* level, so new projects inherit it without a utility module.

5.3 Multi-Scale Reproducibility

At small scale (30 independent NSGA-II runs, pop = 20, ngen = 10, steps = 150), ecological HV = 7.665 (std 3.518, 95% CI [6.424, 8.914]). The distribution is bimodal—runs settle near local optima (HV $\approx$ 4–6.5) or reach the global Pareto front (HV $\approx$ 11.7–11.8). Crucially, at $K = 120$ (original parameterization), the predator went extinct in every episode, reducing both objectives to degenerate constants (HV std = 0.000). The collapsed CI revealed the degeneracy; a single-run study would have reported one number without detecting it. The bimodal convergence (roughly half of runs reaching each mode) is consistent with random initialization placing populations in different basins of attraction; warm restarts from the local-optima mode could shift runs toward the global front, but landscape navigation is orthogonal to the framework contribution.

The study was then replicated at operationally relevant scale (steps = 1,000, n_eval = 5 episodes per genome evaluation, pop = 30, ngen = 15, $n = 20$ runs). At this horizon NSGA-II's structural advantage over random search emerges clearly: mean HV = 16.66 (std 10.84, CI [12.37, 20.94]) versus random mean = 6.00 (std 0.34, CI [5.85, 6.14]). Wilcoxon test: $p = 1.91 \times 10^{-6}$; Cohen's $d = 1.39$ (large effect); NSGA-II achieves 178% higher mean HV than random search. At 150-step scale the same comparison is not significant ($d = 0.49$, $p = 0.49$); the Pareto landscape becomes meaningfully richer only at the longer horizon.

Figure 1 shows per-run distributions and aggregate statistics. The HEAS framework required no code changes between the 150-step and 1,000-step studies: only the `steps` argument changed.

5.4 Tournament Validation and OOD Champion Robustness

Voting-rule agreement and sample complexity were assessed across 30 repeats of the 8-scenario tournament with 100 episodes per scenario. The three participants were the NSGA-II-evolved champion (risk = 0.003, dispersal = 0.959), a reference policy (risk = 0.55, dispersal = 0.35), and a contrarian (risk = 0.85, dispersal = 0.15). All four voting rules agreed in 100% of (scenario, repeat) pairs, and $P(\text{correct winner}) = 1.0$ at all tested episode budgets (4, 10, 25, 50, 100 per scenario). This unanimity confirms genuine dominance. A near-tie sensitivity study (S9) searched for traits maximizing rule disagreement; the most discordant case found (participants with risk $\approx$ 0.62–0.67, dispersal $\approx$ 0.48–0.53) achieves only 75.5% mean off-diagonal agreement (24.5% disagreement), confirming rules diverge on genuinely close contests and that 100% agreement reflects real dominance.

Noise stability was assessed by injecting Gaussian noise $\mathcal{N}(0, \sigma^2)$ into episode scores (30 repeats) and measuring Kendall's τ (Kendall 1938) between clean and noisy rankings. Rankings are stable ($\tau = 1.0$) for noise up to 0.65% of the inter-policy margin, and degrade gracefully beyond. Quantitatively, $\tau = 0.944$ at $\sigma = 10$ (6.5% of margin), $\tau = 0.744$ at $\sigma = 50$, and $\tau = 0.508$ at $\sigma = 200$. Realistic simulation variance is far below 10% of typical inter-policy margins, placing well-designed tournaments in the $\tau > 0.94$ regime.

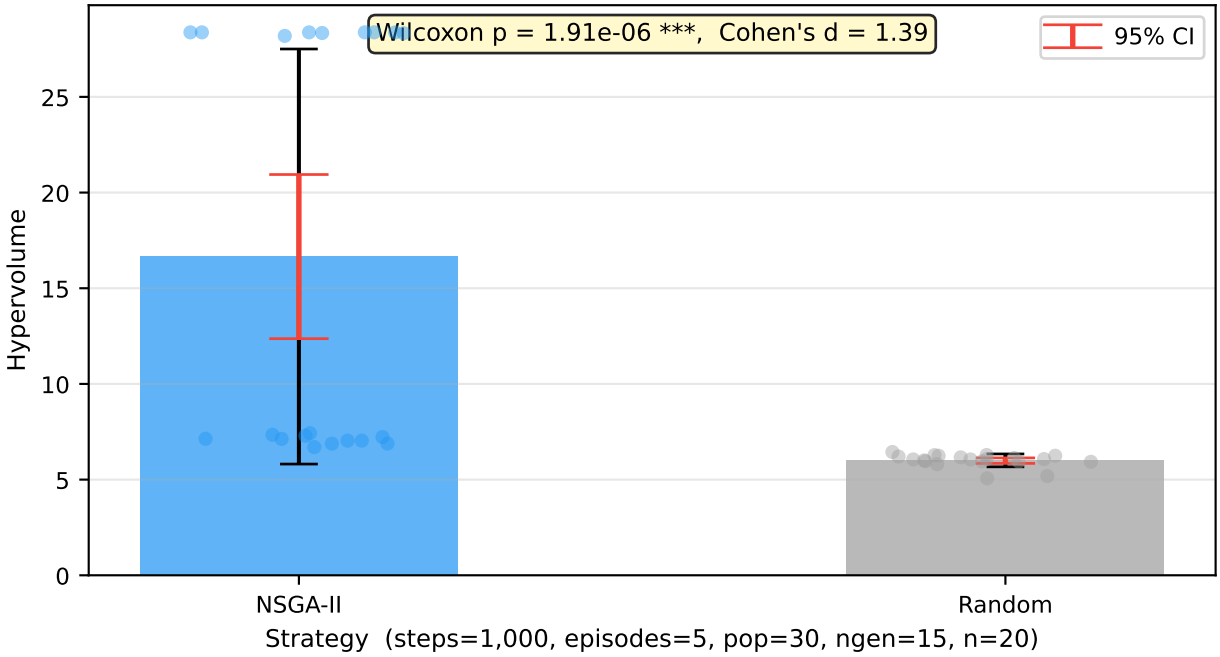

Figure 1: Large-scale algorithm showdown: NSGA-II vs. Random at 1,000 steps/episode ($n = 20$ runs). Bars show mean $\pm$ 1 std; error bars show 95% CI; dots show individual runs. Cohen's $d = 1.39$; Wilcoxon $p = 1.91 \times 10^{-6}$.

Out-of-distribution (OOD) champion robustness was validated by evaluating the evolved ecological champion across held-out scenarios not seen during optimization. At baseline (16 scenarios, steps = 500), the champion won all 16 scenarios (mean $\Delta = +0.88$ biomass, $p < 0.001$, $d = 0.024$); the negligible effect size reflects short evaluation episodes where both policies produce similar terminal biomass, and the consistent win direction is the meaningful generalization signal. At large scale (32 scenarios, steps = 500), the champion genome [0.000107, 0.9335]—low harvest risk, high dispersal—won all 32 scenarios (mean $\Delta = +25.0$ biomass, $+3.96\%$; $p = 4.66 \times 10^{-10}$; $d = 0.17$), with the larger effect size reflecting more informative evaluation episodes.

5.5 Algorithm Agnosticism and Scale Sensitivity

An algorithm ablation at steps = 300 (pop = 20, ngen = 10, $n = 10$ runs) finds that Simple hillclimbing outperforms NSGA-II (means: Simple = 19.66, NSGA-II = 9.99, Random = 7.61). On a 2-gene landscape the Pareto front is effectively one-dimensional; single-objective search concentrates population on the biomass-maximizing direction while NSGA-II distributes across the CV trade-off. This reinforces the agnosticism claim: HEAS allows algorithm selection without any framework changes.

Scale sensitivity is reported in Figure 4, which maps mean HV across a 3×3 grid of steps $\in \{140, 300, 500\}$ and episodes per evaluation $\in \{5, 10, 20\}$ ($n = 10$ runs each). HV rises from 6.0 (140 steps, 5 episodes) to 10.1 (300 steps, 20 episodes) within the small-scale regime, confirming that longer episodes yield richer Pareto landscapes. The bimodal variance pattern persists across all cells, indicating a structural landscape feature rather than a sampling artifact.

The Wolf-Sheep algorithm comparison (steps = 200, pop = 50, ngen = 25, $n = 30$ runs) yields Random = 14.35 (std 0.61), NSGA-II = 10.36 (std 4.07), and Simple = 9.14 (std 3.68). Random achieves the highest HV; a two-sided Wilcoxon rank-sum test confirms the difference is significant ($p = 7.99 \times 10^{-6}$, Random > NSGA-II). The Wolf-Sheep Pareto front consists of a narrow band of sustainable harvest-grazing

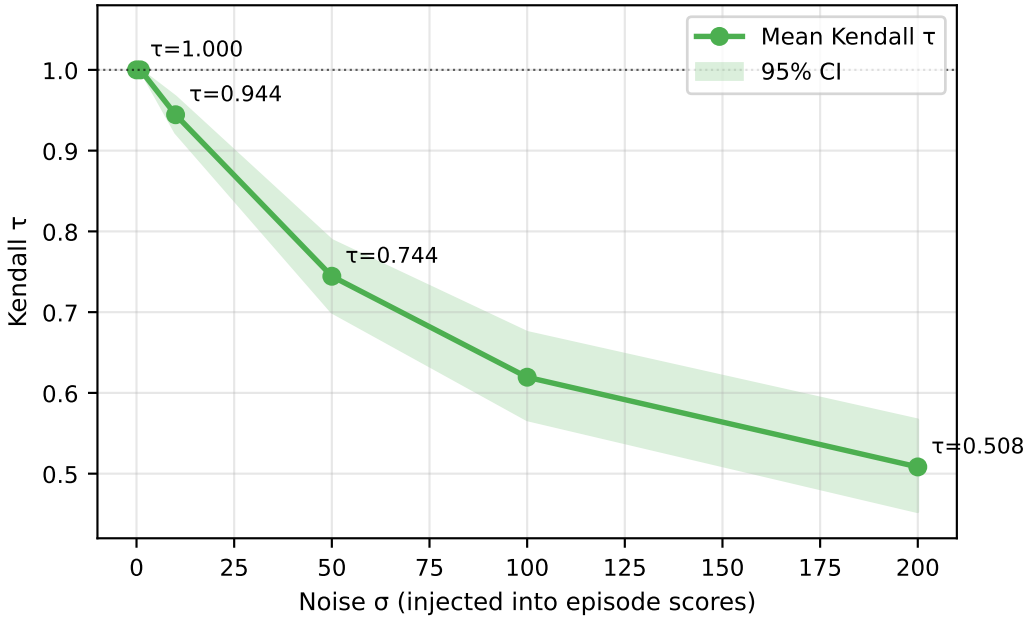

Figure 2: Kendall's τ versus injected score noise σ ($n = 30$ repeats, 8 scenarios). Rankings are stable ($\tau = 1.0$) until noise exceeds $\sim$6% of the inter-policy margin.

combinations; within this concentrated geometry, exhaustive random exploration dominates NSGA-II's diversity-preserving search.

Taken together, the three landscapes yield a landscape-dependent pattern: 2D ecological at small scale (Simple > NSGA-II, effectively one-dimensional front), Wolf-Sheep (Random > NSGA-II, single equilibrium), and 1,000-step ecological (NSGA-II $\gg$ Random, $d = 1.39$, rich multi-objective front). Algorithm-landscape matching—made practical by C3's pluggable optimizer interface—is the operational takeaway: on simple landscapes any search strategy suffices; on complex multi-objective landscapes NSGA-II's structural advantage is large.

5.6 Cross-Domain Portability

This section is a portability demonstration: it asks whether HEAS framework code requires changes when applied to a second domain. The same codebase—unchanged except for the `metrics_episode()` implementation—is run on two heterogeneous domains at large scale ($n = 20$ runs each). In the ecological domain (steps = 1,000), NSGA-II achieves mean HV = 15.46 (std 8.57, CI [12.03, 18.88]); the bimodal convergence pattern persists. In the enterprise domain (steps = 50, 4-gene policy), NSGA-II achieves mean HV = 5.565×10^6 (std 69,624; CI $\pm$ 1.2%), with a unimodal front (high-subsidy, moderate-audit-intensity policies dominate). The small-scale enterprise result (n = 2, Table 2) is insufficient for statistical inference and is a development checkpoint only; a full evaluation equivalent to the ecological study is deferred. The narrow CI ($\pm$1.2%) versus ecological ($\pm$26%) illustrates how domain structure shapes reproducibility. No architectural changes were required: the Arena composition and metric contract remain identical, validating C1 and C2 at operational scale. Tournament validation was also conducted in the enterprise domain across 8 governance scenarios (20 repeats, 30 episodes/scenario) (S10). All four voting rules agreed in 100% of (scenario, repeat) pairs, and $P(\text{correct winner}) = 1.0$ at all tested episode budgets (4, 10, 25, 50), confirming that tournament infrastructure generalizes across domains without framework modification.

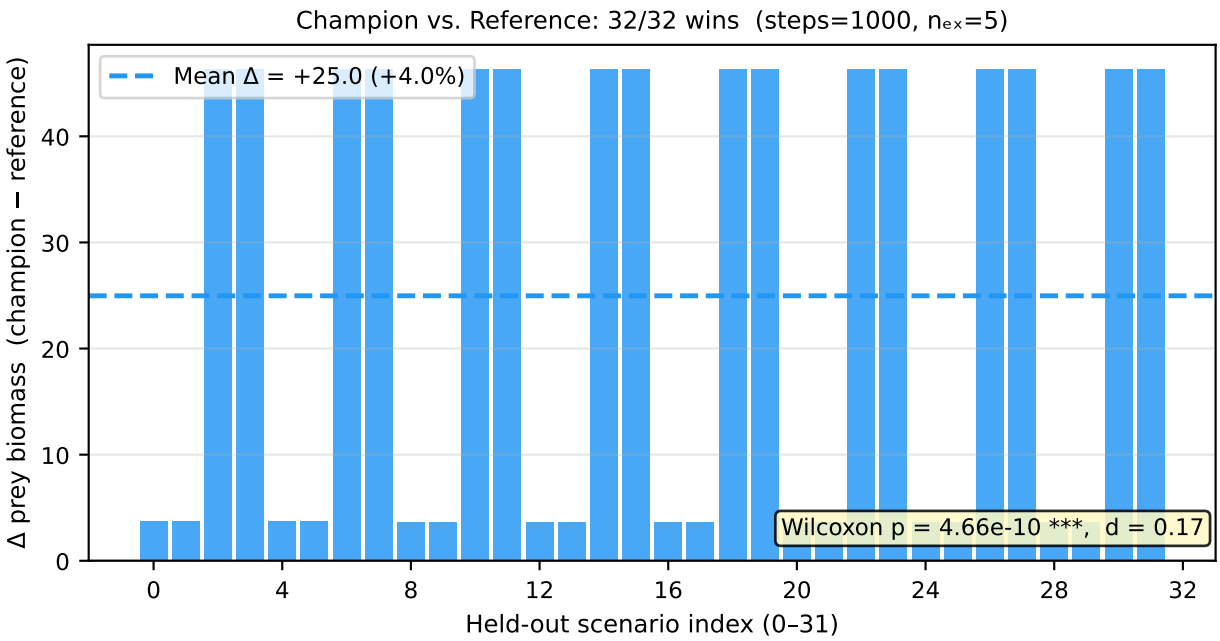

Figure 3: Champion minus Reference biomass across 32 held-out ecological scenarios. Blue bars indicate champion wins (32/32); dashed line marks mean $\Delta = +25.0$ biomass. Wilcoxon $p = 4.66 \times 10^{-10}$.

5.7 Threats to Validity

Construct validity. The case studies are composition vehicles, not calibrated domain models; results are evidence about framework behavior. The metric contract prevents *accidental divergence across consumers*: all three pipeline stages read the same key. An incorrect `metrics_episode()` implementation will be consistently wrong rather than silently inconsistent. The framework additionally exposes `validate_metrics_episode()`, which raises `TypeError` when a `metrics_episode()` implementation returns a non-dict or non-float value, catching contract violations early in development.

Internal validity. Fixed evaluation seeds isolate framework effects but can suppress variance in some metrics. Zero-variance findings are treated as diagnostic signals (e.g., the $K = 120$ degeneracy) rather than performance claims. The champion genome was evolved at steps = 200; OOD validation at steps = 500 (32 scenarios) confirms generalization beyond the training horizon.

External validity. The Mesa baseline is a competent single-project implementation, not a census of all coding styles. Runtime findings for `n_jobs` depend on episode cost and hardware. All three case studies use ODE-style or tabular dynamics; researchers working with genuinely spatially-explicit ABMs—heterogeneous agents on a grid, emergent network effects, rich move-and-interact loops—will encounter integration patterns outside HEAS's current design scope. The 97% LOC reduction is validated specifically for the tabular policy-search workflow that HEAS targets.

6 CONCLUSION

HEAS addresses the coupling code problem through four architectural contributions—Layer/Stream/Arena composition, a uniform metric contract, native NSGA-II integration, and tournament evaluation primitives. Across five evaluation questions, results confirm the framework hypothesis: 97% coupling code reduction (160 vs. 5 LOC), reproducible multi-run behavior at two simulation scales, stable tournament rankings under realistic noise ($\tau > 0.94$), and algorithm flexibility when landscape structure favors simpler optimizers. Three case studies—including a zero-coupling-code mean-field reimplementation of the published Mesa

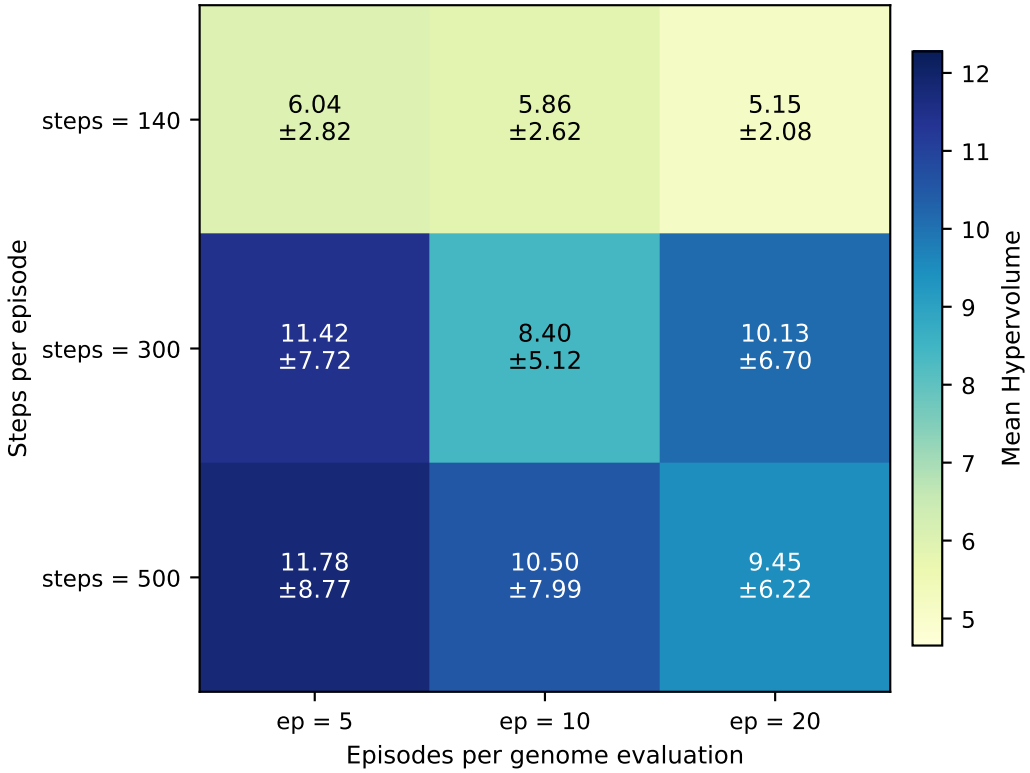

Figure 4: Scale sensitivity heatmap: mean HV by simulation horizon (steps × episodes/genome). Each cell is $n = 10$ independent NSGA-II runs. HV generally increases with steps; non-monotonicities between the 300- and 500-step rows reflect high between-run variance. Episode count has a weaker secondary effect.

Wolf-Sheep model—and a 32-scenario OOD champion stress test ($p < 10^{-9}$) provide broad empirical coverage.

Limitations. For lightweight ODE simulations (<0.01 s/episode), process startup overhead (≈9 s) dominates; use `n_jobs=1`. HEAS is designed for ODE-style and tabular simulations; spatially-explicit ABMs remain Mesa's domain. The metric contract is scalar-focused (`dict[str, float]`); workflows needing agent-level distributions require an additional aggregation layer. Adopting HEAS requires learning the Layer/Stream/Arena abstraction; researchers who need Mesa's browser-based visualization for exploratory single-run analysis should continue using Mesa for that phase and adopt HEAS for headless batch optimization.

Future work. Priorities are a `MesaLayerAdapter` (wrapping existing Mesa models as HEAS Layers without rewriting dynamics), scalable parallel backends (Ray/Dask), and automated scenario generation.

For double-blind review, repository access details are omitted. An anonymized artifact package containing all scripts under `experiments/` is prepared for reviewers.

A REPORTED MEANS AND SOURCES

Source IDs (paths under `experiments/results/`):

S1 mesa_vs_heas/loc_comparison.json
S2 eco_stats/pop20_ngen10_trait/summary.json
S2b ent_stats/summary.json
S3 tournament_stress/noise_stability.json
S4 baseline_comparison/ablation.json

Table 2: Principal means reported in this paper with source file IDs (S1–S8).

Quantity	Value	Source
Coupling LOC reduction	97% (160 vs. 5)	S1
Ecological HV mean (small scale, $n=30$)	7.665 ± 3.518	S2
Enterprise HV mean (small scale, $n=2$)	3762.99	S2b
Noise τ at $\sigma=10$	0.944	S3
Noise τ at $\sigma=50$	0.744	S3
Simple ablation HV mean	19.66	S4
NSGA-II ablation HV mean	9.99	S4
Random ablation HV mean	7.61	S4
Large-scale NSGA-II HV (1,000 steps, $n=20$)	16.66 ± 10.84	S5
Large-scale Random HV (1,000 steps, $n=20$)	6.00 ± 0.34	S5
NSGA-II vs. Random Cohen's d (large scale)	1.39	S5
16-scenario OOD wins / Cohen's d	16/16 / 0.024	S6a
32-scenario OOD wins / Cohen's d	32/32 / 0.17	S6b
32-scenario mean Δ biomass	+25.0 (+3.96%)	S6b
Wolf-Sheep NSGA-II vs. Random p-value	7.99×10^{-6}	S7
Enterprise HV mean (large scale, $n=20$)	5.565×10^{6}	S8
Near-tie mean off-diagonal agreement	0.755 (24.5% disagree)	S9
Enterprise tournament agreement (100%)	100% / $P=1.0$	S10

S5 large_scale_comparison/p1_summary.json
S6a baseline_comparison/champion_vs_ref.json
S6b large_scale_comparison/p3_summary.json
S7 wolf_sheep_study/algo_comparison.json
S8 large_scale_comparison/p2_summary.json
S9 near_tie_tournament/best_case.json
S10 enterprise_tournament/summary.json

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